

BM25 Search

The screenshot shows the RAG Pipeline interface in a web browser. The left sidebar contains a menu with 'SEARCH ENGINES' and 'AI PROCESSING' sections. Under 'SEARCH ENGINES', 'BM25 Search' is selected. The main content area is titled 'BM25 Search' and features a 'Keyword-based search algorithm' button. A search form is present with a 'Search Query' field containing 'User Story: HC-287 - Dynan', a 'Results Limit' field set to '10', and a red 'Search' button. Below the search form is a 'Show Filters' dropdown. The browser's address bar shows 'localhost:3000'.

The screenshot shows the RAG Pipeline interface displaying search results. The left sidebar is identical to the previous screenshot. The main content area shows the 'BM25 Search Results' section, which states 'Found 10 test cases in 146ms' and 'Search Type: bm25'. Below this, a 'Search Results (10)' section is visible, showing the first result: '#1 TC_1199' with a 'Score: 373.87'. The result details include 'Search and attach the form', 'Module: IP digital', 'P2', 'Risk: Low', and 'Automated'. A green banner at the bottom right of the results section says 'Found 10 results in 146ms'. The browser's address bar shows 'localhost:3000'.

RAG Pipeline | Testleaf Enterprise v2.0 • Testleaf

Cluster0 Data | Cloud: MongoDB

Ask Gemini

localhost:3000

BM25 Search

Keyword-based search algorithm

#1 TC_1199 Score: 373.87

Search and attach the form

Module: IP digital P2 Risk: Low Automated

Description: to search and attach the form

Steps: 1.Launch the application 2.Login to the application with valid credentials. 3.Landing screen - should be able to view multiple tabs like My patients, all patients, new patients,discharge plan,tasklist ,drug administration , drug inventory, handover takeover & patient check out 5.Select on patient label 6.Click on form 7.Select search and attach the form 8.Search the form 9.Enter the details 10.Save and lock the form

Expected Results: To search and attach the form to the patient

Created: 8/17/2026

#2 TC_3921 Score: 372.86

Radiology-Transactions-Patient Queue Dashboard

Module: Radiology P3 Risk: Low Manual

Description: Verify user is able to View Patient Queue Dashboard

Steps: 1.Launch the Application. 2.Login to the application with valid credentials 3.Click on Patient Service Tab and select Radiology from the drop down. 4.Navigate to Module ->Diagnostics and Imaging ->Transactions ->Patient Queue Dashboard 5. Select Modality and Click Go

Expected Results: Patient Queue Details Should be Displayed

Vector Search

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Cluster0 Data | Cloud: MongoDB

Ask Gemini

localhost:3000

Vector Search

Semantic AI vector search

Query Search

Search through test cases using semantic vector embeddings.

Search Query

Enter your search terms and find similar test cases

Enter your search query

Share Diagnostic Reports with Patients via WhatsApp

Clear

Use descriptive terms related to test functionality

TRY AN EXAMPLE QUERY

Share Diagnostic Reports with Patients via... Login test cases for the Billing module wi... Validate payment gateway timeout and retry... Password reset flow across web and mobile High-risk regression scenarios for checkou...

Advanced Metadata Filters Show Filters

Results Limit 5 Results

Search

RAG Pipeline
Enterprise v2.0 • Testleaf

Query Preprocessing

Vector Search

BM25 Search

Hybrid Search

Score Fusion

AI PROCESSING

Summarize & Dedup

Prompt & Schema

Enterprise AI Stack
MongoDB Atlas • Mistral AI • Groq

RAG Pipeline > Vector Search

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Advanced Metadata Filters

Show Filters

Results Limit

5 Results

Found 5 results

RAG Pipeline
Enterprise v2.0 • Testleaf

Query Preprocessing

Vector Search

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RAG Pipeline > Vector Search

Semantic AI vector search

5 results ranked by semantic similarity

Test Case	Similarity	Description	Source
CONSULTANT DASHBOARD - M	90.3%	Verify user able to Save Diagnosis	
Doctors-Wards			
REPORTS - CLINICAL REPORTS	90.2%	Verify user able to view Clinical Report	
Doctors-Wards			
REPORTS - CLINICAL REPORTS	90.1%	Verify user able to view Clinical Report	
Doctors-Wards			
DOCTOR DASHBOARD - OP DA	90.1%	Verify user able to Save Diagnosis	
Doctors-Wards			

Rows per page: 10 1-5 of 5

Detailed View

Consultant Dashboard - My Appointments - - OP Dashboard -

Hybrid Search

The screenshot displays the 'Hybrid Search' configuration page in the RAG Pipeline interface. The left sidebar contains navigation options: Vector Search, BM25 Search, Hybrid Search (selected), Score Fusion, Summarize & Dedup, and Prompt & Schema. The main panel shows the 'Hybrid Search' settings, which combine BM25 keyword matching with semantic vector search. The search query is 'Share Diagnostic Reports wi', and the results limit is set to 10. The search weights are configured as follows: BM25 Weight (Keyword) at 30% and Vector Weight (Semantic) at 70%. Below the sliders, there are three preset buttons: 'Balanced (50/50)', 'Keyword Heavy (70/30)', and 'Semantic Heavy (30/70)'. The 'Show Filters' button is also visible.

RAG Pipeline | Testleaf Enterprise v2.0 • Testleaf

Cluster0 Data | Cloud: Mongo0

localhost:3000

Ask Gemini

RAG Pipeline > Hybrid Search

Combined BM25 + Vector search

Hybrid Search
Combines BM25 keyword matching with semantic vector search for best results.

Search Query: Share Diagnostic Reports wi Results Limit: 10 Search

Search Weights

BM25 Weight (Keyword): 30% Vector Weight (Semantic): 70%

0% 50% 100% 0% 50% 100%

Balanced (50/50) Keyword Heavy (70/30) Semantic Heavy (30/70)

Show Filters

Generate TC with RAG

The screenshot displays the 'Generated Test Cases with RAG Pipeline' page. The left sidebar shows the navigation options, with 'Prompt & Schema' selected. The main panel shows the 'Generated Test Cases (3)' section. It includes a table with columns: ID, Title, Module, Priority, Type, and Preconditions. The table contains one entry: TC_6355, Positive Flow - View Pending Prescriptions, Verify and Sync to Pharmacy, Prescription Management, P1, Functional. The preconditions are: Doctor user (UHID: D12345) is logged into the hospital portal; patient record (UHID: P67890) has at least one pending prescription generated by another user. The interface also shows the mode 'With RAG', the model 'openai/gpt-oss-120b', and the results '50 MongoDB Matches' and '4,084 / 8000 Tok (51% Budget Safe)'. There is an 'Export CSV' button.

RAG Pipeline | Testleaf Enterprise v2.0 • Testleaf

Cluster0 Data | Cloud: Mongo0

localhost:3000

Ask Gemini

RAG Pipeline > Prompt & Schema

Configure system prompt templates

Generated Test Cases with RAG Pipeline

Mode: With RAG Model: openai/gpt-oss-120b 50 MongoDB Matches 4,084 / 8000 Tok (51% Budget Safe)

Export CSV

Generated Test Cases (3) Retrieved MongoDB Context (7) Linked User Stories (10)

ID	Title	Module	Priority	Type	Preconditions
TC_6355	Positive Flow - View Pending Prescriptions, Verify and Sync to Pharmacy	Prescription Management	P1	Functional	Doctor user (UHID: D12345) is logged into the hospital portal; patient record (UHID: P67890) has at least one pending prescription generated by another user

Generate TC without RAG

The screenshot shows the 'RAG Pipeline' interface on a web browser. The left sidebar contains navigation options: 'Vector Search', 'BM25 Search', 'Hybrid Search', 'Score Fusion', 'Summarize & Dedup', and 'Prompt & Schema'. The main content area is titled 'Generated Test Cases Without RAG (Direct Baseline)'. It displays the following information:

- Mode: Without RAG
- Model: openai/gpt-oss-120b
- 0 DB Retrieval (Zero Context)
- 3,517 / 8000 Tok (44% Budget Safe)
- Export CSV button

Below this information is a table with the following columns: ID, Title, Module, Priority, Type, and Precond. The table is currently empty.

Comparison

The screenshot shows the 'RAG Pipeline' interface on a web browser. The left sidebar is the same as in the previous screenshot. The main content area is titled 'AI Comparative Analysis & Pedagogical Evaluation'. It displays the following information:

- Multi-LLM Evaluated
- Checkmark icon: RAG anchors test generation in real hospital data, slashing hallucination and boosting relevance
- Text: Approach B pulls concrete prescription-module artifacts from the MongoDB knowledge base, while Approach A fabricates generic steps from thin air. The RAG-enhanced output stays tied to actual entity names, database fields, and compliance checks, making the tests far more executable in an enterprise healthcare setting.
- COMPARATIVE DIMENSION SCORING
- Domain Grounding & Healthcare Entities: Without RAG: 45% | With RAG: 96%
- Text: Approach A invents UHIDs, module names and UI labels without any reference to the hospital's data dictionary, whereas Approach B re-uses real identifiers (e.g., doctor UHID D77777, patient UHID P99999) and respects domain-specific concepts such as PHI, consent flags, and encrypted PDF links that appear in the source schema.
- Database & Architecture Alignment: Without RAG: 30% | With RAG: 95%
- Text: The baseline test cases assume a generic web portal and ignore the HMS micro-service contracts, audit-log tables, and the WhatsApp integration layer. The RAG-driven cases cite actual MongoDB test-case IDs (TC_1140, TC_1193, TC_4406) and mirror the real prescription verification workflow, demonstrating alignment with the existing architecture.

The screenshot displays the 'RAG Pipeline v2.0' interface within the Testleaf application. The main content area is titled 'Prompt & Schema' and features three distinct configuration sections, each with a 'Configure system prompt templates' button.

- Workflow & Verification Depth:** This section describes high-level actions (login, approve) while omitting concrete verification artifacts like toast messages, audit-log entries, or pagination counts. It specifies observable UI cues, audit-log checks, and pagination behavior, providing clear pass/fail criteria that can be automated. Deeper verification points enable deterministic test automation and faster defect isolation.
- Database Grounding:** This section notes the absence of reference to underlying collections, fields, or existing test-case IDs, as steps are built on assumed generic tables. It cites real MongoDB test-case IDs (TC_1140, TC_1193, TC_4406) and mirrors the actual schema used by the HMS, ensuring that generated tests are executable against the production-like environment. Grounded tests reduce false positives/negatives caused by mismatched data structures and accelerate integration testing cycles.
- Key Teaching Points for Learners & QA Architects:** This section highlights that RAG injects factual, system-specific context that transforms vague LLM output into actionable test scripts.

The left sidebar provides navigation options for various AI processing tasks, including 'Summarize & Dedup', 'Prompt & Schema' (currently selected), and 'Enterprise AI Stack' (MongoDB Atlas, Mistral AI, Groq). The top navigation bar includes links to 'Ask Gemini', 'Cluster0 Data | Cloud: MongoD...', and 'localhost:3000'.

